CS6493: Natural Language Processing - Assignment 2

Instructions

1. Due at 10 pm, April 29, 2026;
2. You can submit your answers by **a single PDF with the code package** or **a single Jupiter notebook** containing both the answers and the code;
3. For the coding questions, besides the code, you are encouraged to additionally give some descriptions of your code design and its workflow. Detailed analysis of the experimental results is also preferred;
4. Total marks are 100;
5. If you have any questions, please post your questions on the Canvas-Discussion forum or contact TA Mr. Junyi ZHOU (junyizhou8-c@my.cityu.edu.hk).

Question 1

(**20 marks**) Machine translation in natural language processing (NLP) refers to the task of automatically translating text or speech from one language to another using computational methods. It involves developing algorithms and models that can understand the meaning of the source language and generate an equivalent translation in the target language.There are several approaches to machine translation in NLP, including rule-based methods, statistical models, and neural machine translation (NMT) models.

1. The model needs to use matrices to accelerate operations, but training requires data to be input into the model in batches. The sentences in a batch need to be of equal length, so padding is required. Padding is the process of filling sequences with 0 to make them the same length.

   Sentences:

   - Sentence 1: I like cats and dogs
   - Sentence 2: She enjoys reading books and playing video games
   - Sentence 3: They play football every weekend

   vocabulary (token → index):

I	4
She	5
They	6
like	7
enjoys	8
play	9
cats	10
dogs	11
reading	12
books	13
football	14
every	15
weekend	16
and	17
playing	18
video	19
games	20

   (a) Tokenize each sentence and convert it into a sequence of indices using the vocabulary above. **(2 marks)**

   (b) Pad all sequences on the right with 0 so that all sequences have the same length as the longest sentence **(2 marks)**

2. In neural NLP models, words are often converted into vectors before being used as input.

   (a) What is a word embedding? **(2 marks)**

   (b) Explain one advantage of word embeddings compared with one-hot encoding. **(2 marks)**

3. Beam search and greedy search are two common decoding algorithms used in sequence generation tasks such as machine translation or text generation.

   Figure 1 shows a decoding tree, where each edge represents the probability of generating the next token.

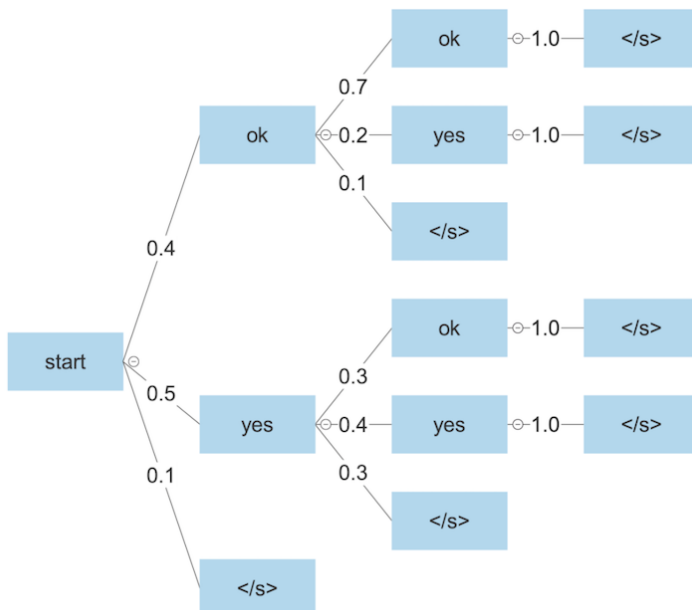

Figure 1: Decoding search tree

   Answer the following questions based on the decoding tree in Figure 1:

   (a) Using greedy search, find the generated sequence and compute its probability. **(2 marks)**

   (b) Using beam search with beam width $k = 2$, find the most probable sequence and show the intermediate steps. **(2 marks)**

   (c) Briefly explain why beam search can produce better results than greedy search. **(2 marks)**

4. BLEU (Bilingual Evaluation Understudy) is a popular evaluation metric used in machine translation to assess the quality of translated text.

   BLEU Formula:

$$\text{BLEU} = BP \cdot \exp\left(\sum_{n=1}^{4} w_n \ln P_n\right)$$

Candidate sentence:
A small cat is sitting on the wooden table.

Reference sentence 1:
A little cat is sitting on the leather chair.

Reference sentence 2:
A small cat is sitting on a wooden chair.

Assume uniform weights w_n. Compute the BLEU score with $N = 4$. Show your calculation steps**(6 marks)**

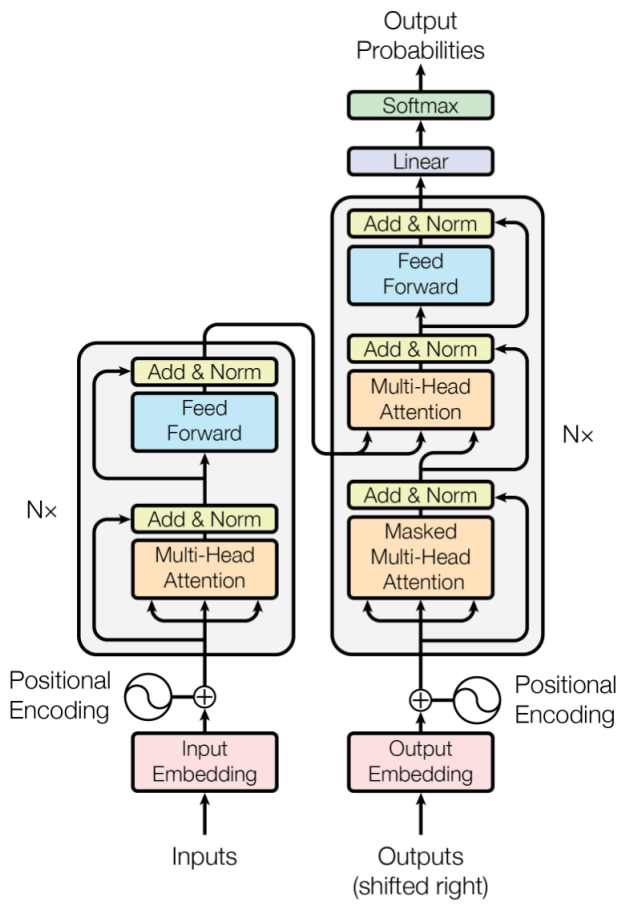

Figure 2: Model architecture of the Transformer.

Question 2

(**20 marks**) Traditional sequence-to-sequence models for tasks such as machine translation commonly rely on RNN/LSTM encoder-decoder architecture. However, their sequential computation limits parallelism and makes it difficult to model long-range dependencies effectively. Vaswani et al. [3] proposed the **Transformer**, a model architecture that relies entirely on attention mechanism, removing recurrence and convolution from the core sequence modeling components. The Transformer is built from stacked encoder and decoder blocks, where attention is the central operation. A core component is **Scaled Dot-Product Attention**. Given a query matrix Q, a key matrix K, and a value matrix V, the attention operation is defined as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V,$$

where d_k is the dimensionality of the keys (and queries). Intuitively, for each query, the model computes similarity scores against all keys, normalizes them with softmax into attention weights, and then forms a weighted sum of the values.

1. In self-attention, we typically for $Q = XW^Q$, $K = XW^K$, and $V = XW^V$ from the same input $X \in \mathbb{R}^{n \times d_{model}}$. But suppose someone propose to simplify parameters by setting $W^Q = W^K$

and even further $V = K$. Under **only** the constraint $W^Q = W^K$, what property must the attention score matrix $S = QK^\top$ satisfy and how would that property limit the model's ability to represent directional or asymmetric relations? And if we additionally force $V = K$, can you give a brief explanation of what expressivity is lost by conflating keys with values? (**8 marks**)

2. Consider a single-query attention setting with a query vector $q \in \mathbb{R}^{d_k}$, a set of key vectors $\{k_i\}_{i=1}^n$ with $k_i \in \mathbb{R}^{d_k}$, and a set of value vectors $\{v_i\}_{i=1}^n$ (with compatible dimensionality). Define attention weights and the attention output as:

$$\alpha_i = \frac{\exp(\beta q^\top k_i)}{\sum_{j=1}^n \exp(\beta q^\top k_j)}, c = \sum_{i=1}^n \alpha_i v_i,$$

where $\beta > 0$ is a scaling coefficient applied to the dop-product scores (i.e., it rescales the logits before softmax, and in particular, setting $\beta = \frac{1}{\sqrt{d_k}}$ recovers the Transformer's scaled dot-product attention in its original formulation [3]). Describe what distribution $\{\alpha_i\}$ approaches as $\beta \to 0$ and briefly state the corresponding effect on c; and what if $\beta \to \infty$? (**7 marks**)

3. In the decoder's **masked self-attention**, let $Q, K, V \in \mathbb{R}^{n \times d_k}$. The attention is defined as:

$$A = \text{softmax}\left(\frac{QK^\top + M}{\sqrt{d_k}}\right), O = AV,$$

where $M \in \mathbb{R}^{n \times n}$ is a **causal mask** used to enforce autoregression, which means position t is not allowed to attend to any future position $j > t$. Explain why the mask must be added before the softmax rather than applying it after the soft max by zeroing out some attention weights. Finally, for $n = 4$, write down the explicit 4×4 mask matrix M using entries 0 and $-\infty$. (**5 marks**)

Question 3

(**30 marks**) This question is based on the lecture materials on understanding tasks (NLU) and generation tasks (NLG).

1. NLU focuses on comprehending and extracting meaning from language, while NLG focuses on generating human-like text. In your own words, state the key difference between NLU and NLG and give two concrete task examples for each category. (**4 marks**)

2. For the following text classification applications, identify the most appropriate classification type for each (e.g., binary, single-label multi-class, etc.):

   (a) Decide whether an email is spam or legitimate. (**2 marks**)

   (b) Assign a news article to exactly one section from {HomeNews, International, Entertainment, Lifestyle, Sports}. (**2 marks**)

   (c) Tag a technical paper with all applicable category labels in a taxonomy. (**2 marks**)

   (d) Predict a review rating from an ordered set {Disastrous, Poor, Mediocre, Good, Excellent}. (**2 marks**)

3. Reading comprehension (RC) can be framed as answering questions given a passage, and answers may be extractive or abstractive:

   (a) In one sentence, explain what extractive QA means in the context of SQuAD-style [2] RC. (**2 marks**)

   (b) Give two reasons why SQuAD was considered better than some earlier QA datasets. (**4 marks**)

   (c) SQuAD 2.0 [1] contains many unanswerable questions; what additional capability must a system have to perform well on SQuAD 2.0? (**2 marks**)

4. For sequence generation (e.g., machine translation), common decoding strategies include greedy decoding and beam search. BLEU is a widely used automatic evaluation metric for machine translation:

   (a) What is greedy decoding, and what is its main drawback? (**2 marks**)

   (b) Why is exhaustive search decoding impractical? (**2 marks**)

   (c) Briefly explain beam search and what the beam size k controls; also explain why length normalization to commonly used. (**2 marks**)

   (d) BLEU uses n-gram precision and a brevity penalty; explain whay each component is intended to discourage. (**2 marks**)

5. Human evaluation is often expensive or slow, so practical systems frequently use heuristic automatic metrics for generation quality. Name two such heuristic metrics and state whay each is roughly trying to capture. (**2 marks**)

Question 4

(30 marks) Large language models and question answering systems are increasingly used in real-world applications such as education, customer service, and internal knowledge management. In this question, you will design and analyze practical NLP systems.

1. **Designing a Campus Assistant System (15 marks)**

   A university wants to deploy an AI assistant that answers students' questions about course registration, deadlines, grading policies, and graduation requirements. The information is stored in PDFs, web pages, and policy documents that are updated every semester.

   The university has two main requirements:

   - Answers must be based on official documents.
   - The system should reduce the risk of giving incorrect or outdated information.

   (a) Propose a system design to solve this problem. Clearly describe the main components (for example, document storage, retrieval, language model, or ranking). (8 marks)

   (b) Describe the step-by-step workflow from a student asking a question to the system producing an answer. (7 marks)

2. **Handling Uncertain or Missing Information (15 marks)**

   After deployment, the university discovers that students sometimes ask questions that are:

   - Not covered in the documents, or
   - Ambiguous or poorly phrased.

   The current system still tries to answer and occasionally produces incorrect responses.

   (a) Explain why modern language models may still produce answers even when they are uncertain. (6 marks)

   (b) Propose two practical strategies to make the system safer and more reliable in these situations. Explain how each strategy helps. (9 marks)

References

[1] P. Rajpurkar, R. Jia, and P. Liang. Know what you don't know: Unanswerable questions for squad. *arXiv preprint arXiv:1806.03822*, 2018.

[2] P. Rajpurkar, J. Zhang, K. Lopyrev, and P. Liang. Squad: 100,000+ questions for machine comprehension of text. *arXiv preprint arXiv:1606.05250*, 2016.

[3] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 30, 2017.